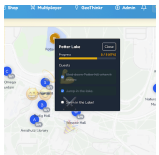
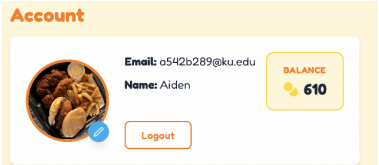
Sprint 1 Requirement Artifacts

eecs582-sidequest.vercel.app
https://github.com/TheOfficialBarn/eecs582-sidequest/
Sprint artifacts located in the repo: *documentation/582/sprint1*

1. Add account points/currency to DB **(done)**
   - Verify DB table + API read/write path
   - New column/field exists for points. New user should have 0 points.
   - Points update globally after refresh
   - Screenshot of DB schema with points column added to User table (***included below***)
2. Add points animation **(done)**
   - When points are awarded, animation triggers exactly once per award event
   - Verify the Event + UI when points are awarded
   - Image of successful award → animation

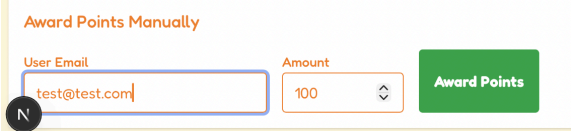


3. Add points counter to user page **(done)**
   - user page needs to show points (verify User/Profile page)
   - refreshing page shows persistent values
   - Screenshot of profile page with correct points visible

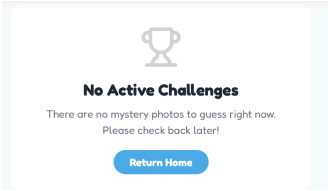


4. Add points counter to toolbar **(done)**
   - Verify point counter on toolbar, and verify it on global nav bar/tool bar
   - Points are visible on every page where toolbar appears
   - No overflow display on large values (999+ points)
   - Screenshot of toolbar showing functioning points

5. Add admin tool to add points **(done)**
   - Verify admin page + permissions
   - Admin can select a user + enter amount + submit
   - User's points update in DB and reflect in UI
   - Screenshot of admin adding points and user points increasing

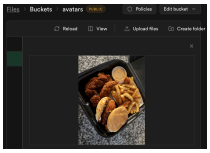


6. Award points for completing quests **(done)**
   - Verifying the question completion lifecycle
   - Completing a quest triggers points award
   - Same quest cannot be exploited repeatedly
   - Screen shot of empty quest page, proving no exploit available

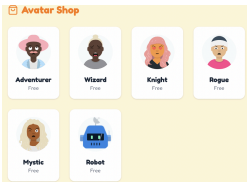


7. Add profile pictures functionality **(done)**
   - Screenshot: profile picture displayed
   - users can set/change profile pictures
   - Selected picture persists after refresh/re-login

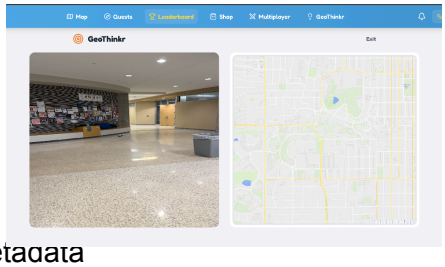
8. Create initial set of profile pictures **(done)**
   - verify initial set in folder
   - At least a number of profile picture are set
   - Screenshot of the Supbase directory containing profile pictures



9. Add tab for profile picture shop **(done)**
   - verify on nav bar + shop page



- Shop displays available pictures + price
- Screenshot/video of a “purchase/select” action and resulting change
- Screenshot of shop page functioning successfully



10. Create Tab for Campus Guesser game **(done)**
    - Nav tab + route to Campus Guesser landing screen
    - screenshot of tab + landing screen.

11. Create database of KU campus photos with correct location metadata
    - DB table schema + sample rows (photo id/url + lat/long + tags + difficulty)

12. Build game interface displaying random campus photo
    - Interface that displayed random photos of campus via JPEG, PNG, etc.
    - Gameplay screen showing random photo render; short video demonstration

13. Create interactive KU campus map for user guessing
    - interactive and dynamic map, 1:1 to KU campus
    - Map UI with clickable guess placement; screenshot showing pin drop + coordinates captured
14. Implement scoring system based on guess proximity
    - Scoring formula documented + example calculation (two sample guesses) + UI showing score result
15. Add round timer with configurable time limits
    - Timer visible + settings control + proof it ends round at 0 via short screen recording.

16. Display correct location after each guess with accurate distance info
    - Post-guess results screen: correct pin + user pin + distance label.

17. Create multi-round game structure (5 rounds per game
    - Round counter + transitions; video showing rounds 1→5.

18. Calculate and display total score at end of game
    - End screen with total score + per round breakdown

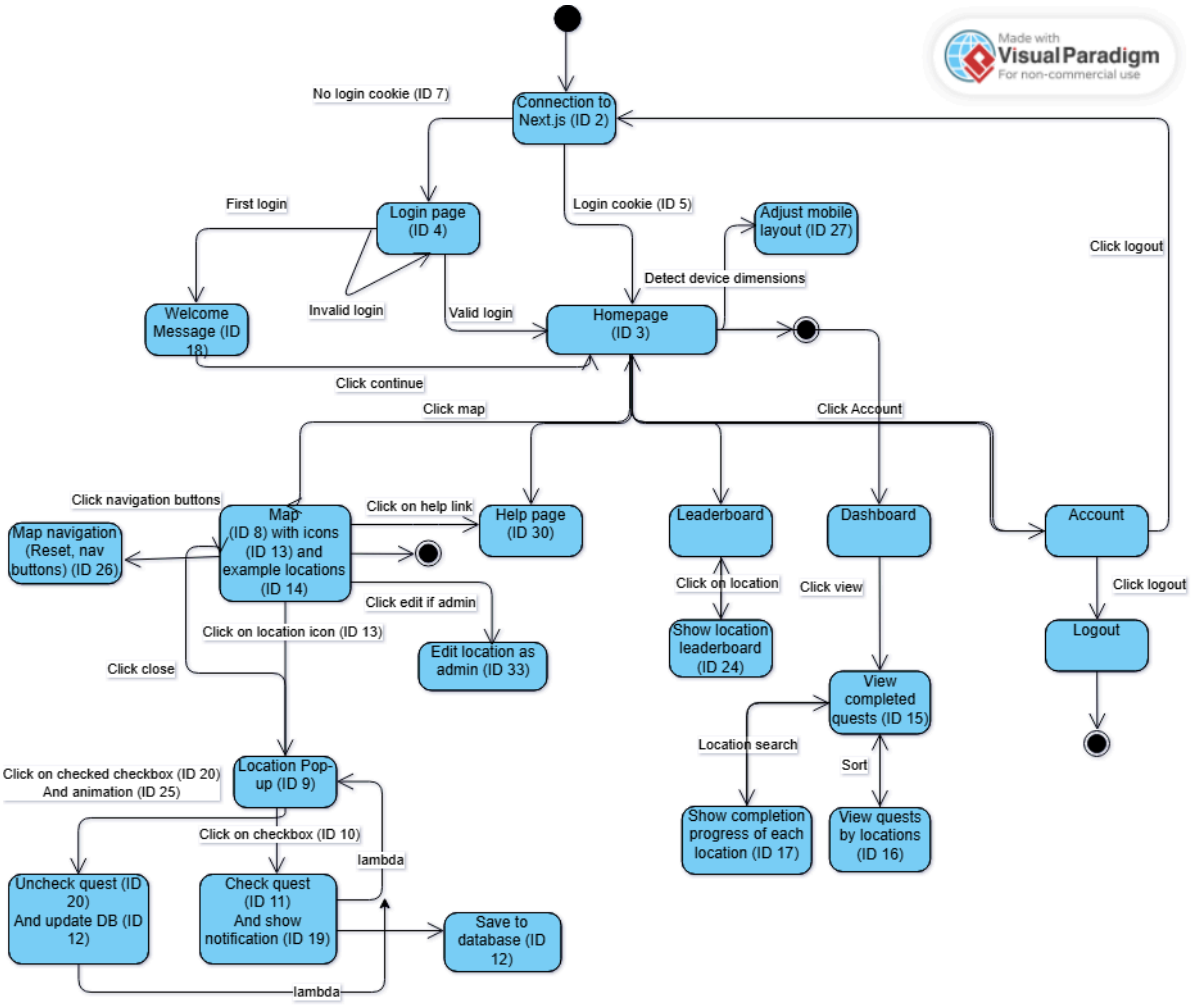
19. Link CampusGuesser scores to existing user account points
    - After game end, points increase in user wallet; screenshot DB + UI.

20. Add CampusGuesser achievements to user profile
    - Achievement earned event + profile display.
    -
21. Update Side Quest homepage to feature CampusGuesser
    - Add CampusGuesser section/card on homepage
    - Add CTA button (“ Play CampusGuesser”)
    - CTA routes to CampusGuesser landing page

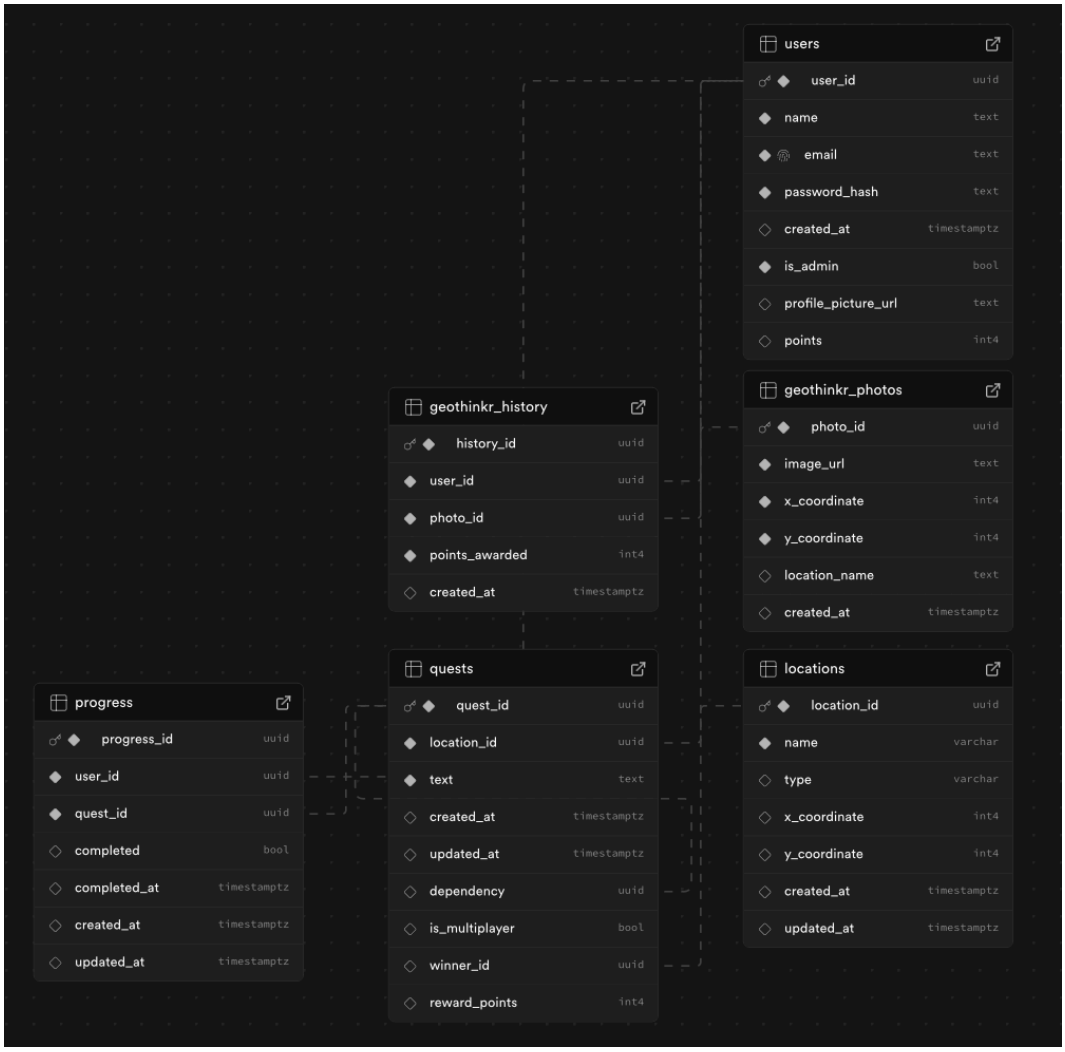
22. Create admin interface for adding new campus photos
    - Admin-only upload form (image and metadata)
    - Submitting creates DB row successfully
    - Non-admin user are blocked from access
23. Implement photo tagging system (buildings, landmarks, etc.)
    - Add tags field to photos in DB
    - Admin can add/edit tags per photo
    - Admin can filter/search photos by tag
24. Add difficulty rating to photos (easy/medium/hard)
    - Add difficulty rating to photos (easy/medium/hard)
    - Admin can set difficulty on upload/edit
    - Game can pull photos by selected difficulty
25. Create photo verification system to ensure accuracy
    - Add status: pending/approved/rejected
    - New uploads default to pending
    - Game only uses approved photos
26. Implement difficulty levels (different campus areas / photo pools)
    - Add difficulty selection before starting
    - Each difficulty uses a different photo pool
    - Different selections produce different photo sets
27. Add hint system (reveal building/general area)
    - Add hint button on gameplay screen
    - Hint reveals building/general area info
    - Add rule: cooldown or point cost
28. Create timed challenge mode with speed bonuses
    - Add mode selector (Normal vs Timed)
    - Timed mode adds countdown + ends at 0
    - Bonus points scale with time remaining
29. Add daily challenge with specific photo set
    - Add “Today’s Challenge” entry point
    - Daily photo set is deterministic for all users
    - Daily set changes based on date (server-defined)
30. Create CampusGuesser leaderboard (daily/weekly/all-time) — Called “GeoThinkr” in app
    - Leaderboard page with tabs (daily/weekly/all-time)
    - Sort by score (desc) with tie-break rule
    - Shows usernames + scores + ranks
31. Add animations for leaderboard records
    - Animate rows on load/new entry
    - Animate rank changes (up/down)
    - Animation triggers once per update
32. Implement share score to social media feature
    - Add “Share” button on results screen
    - Generate share text + link (or image card)
    - Open share modal / copy-to-clipboard

33. Update map detail for CampusGuesser
    - Improve map styling (labels/paths/landmarks)
    - Constrain bounds to KU campus area
    - Keep map performance smooth
34. Design responsive game interface for mobile devices
    - Layout adapts to mobile widths cleanly
    - No overflow for timer/buttons/map
    - Works on at least 2 common viewport sizes
35. Add tutorial/onboarding for first-time players
    - First-run tutorial overlay walkthrough
    - “Don’t show again” toggle
    - Persist tutorial completion state
36. Implement visual feedback for guess accuracy (heatmap/ring)
    - Show accuracy visual after each guess
    - Include legend for meaning of colors/rings
    - Keep distance label visible too
37. Create game statistics page (accuracy, favorite locations)
    - Add stats page linked from profile/game
    - Track 3+ metrics (avg distance, best score, games played)
    - Show “favorite locations/tags” summary
38. Set up Supabase tables for CampusGuesser game data
    - Create tables for games/rounds/guesses/photos
    - Add FKs + indexes for key relationships
    - Store enough data to rebuild results history
39. Create API endpoints for game logic and scoring
    - Endpoints: start game, get photo, submit guess, end game
    - Validate auth + inputs server-side
    - Return consistent JSON + status codes
40. Implement photo randomization algorithm avoiding repeats
    - Use shuffle/bag method per game session
    - Guarantee no repeats within a 5-round game
    - Reset only on new game or bag exhaustion

Requirements Diagram Artifact:



Database Diagram/Schema:



table_name	column_name	data_type	is_nullable	column_default
geothinkr_history	history_id	uuid	NO	gen_random_uuid()
geothinkr_history	user_id	uuid	NO	NULL
geothinkr_history	photo_id	uuid	NO	NULL
geothinkr_history	points_awarded	integer	NO	0
geothinkr_history	created_at	timestamp with time zone	YES	now()
geothinkr_photos	photo_id	uuid	NO	gen_random_uuid()
geothinkr_photos	image_url	text	NO	NULL
geothinkr_photos	x_coordinate	integer	NO	NULL
geothinkr_photos	y_coordinate	integer	NO	NULL
geothinkr_photos	location_name	text	YES	NULL
geothinkr_photos	created_at	timestamp with time zone	YES	now()
locations	location_id	uuid	NO	gen_random_uuid()
locations	name	character varying	NO	NULL
locations	type	character varying	YES	NULL
locations	x_coordinate	integer	YES	NULL
locations	y_coordinate	integer	YES	NULL
locations	created_at	timestamp with time zone	YES	now()
locations	updated_at	timestamp with time zone	YES	now()
progress	progress_id	uuid	NO	gen_random_uuid()
progress	user_id	uuid	NO	NULL
progress	quest_id	uuid	NO	NULL
progress	completed	boolean	YES	false
progress	completed_at	timestamp with time zone	YES	NULL
progress	created_at	timestamp with time zone	YES	now()
progress	updated_at	timestamp with time zone	YES	now()
quests	quest_id	uuid	NO	gen_random_uuid()
quests	location_id	uuid	NO	NULL
quests	text	text	NO	NULL
quests	created_at	timestamp with time zone	YES	now()
quests	updated_at	timestamp with time zone	YES	now()
quests	dependency	uuid	YES	NULL
quests	is_multiplayer	boolean	YES	false
quests	winner_id	uuid	YES	NULL
quests	reward_points	integer	YES	100
users	user_id	uuid	NO	gen_random_uuid()
users	name	text	NO	NULL
users	email	text	NO	NULL
users	password_hash	text	NO	NULL
users	created_at	timestamp with time zone	YES	now()
users	is_admin	boolean	NO	false
users	profile_picture_url	text	YES	NULL
users	points	integer	YES	0